



Hypofibrinolysis and the insulin resistance syndrome

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Atherothrombosis seems now to be the most appropriate term to describe the pathogenic events leading to the development of cardiovascular disease. An impaired fibrinolysis may not only contribute to and aggravate the tendency to thrombosis but fibrin deposit may also play a role in the development of the atherosclerotic plaque. Hypofibrinolysis is observed among obese subjects and it has been shown that an excess of plasminogen activator inhibitor 1 (PAI 1) the main regulator of the fibrinolytic system, is closely associated to other components of the insulin resistance syndrome, namely, excessive body weight, high waist to hip ratio, elevated blood pressure, hyperinsulinemia and hypertriglyceridemia. PAI 1 levels decrease with measures attempting at decreasing insulin resistance.

However, the mechanisms leading to increased PAI 1 levels are still unknown. On the basis of epidemiological studies, and *in vitro* studies with PAI secreting cells such as hepatocytes or endothelial cells, insulin, insulin precursors, and lipoproteins, mainly VLDL, appear candidates for triggering this excessive secretion, but no definite answer has yet been found.

Keywords: insulin resistance; fibrinolysis; plasminogen activator inhibitor; insulin

More than 20 years ago, hypofibrinolysis was shown to be associated with obesity,^{1,2} its degree of impairment being proportional to the body mass index (BMI).¹ Hypofibrinolysis is also associated with hypertriglyceridemia³ and non insulin dependent diabetes.⁴⁻⁷ All these conditions are characterized by resistance to insulin mediated glucose disposal, and at least for the two former by hyperinsulinemia.

The importance of the fibrinolytic system in the development of cardiovascular disease is now well established.^{8,9} Cardiovascular diseases as seen by the clinician, are the consequence of an ischemic event and this event represents the end point of a long pathogenetic process. Atherogenesis leads to formation of atherosclerotic plaque. However the atheroma related morbidity/mortality is due to ulceration followed by subsequent thrombosis which occur on the plaque.¹⁰ The term athero-thrombosis is probably the best one to describe the whole physiopathological process. An impaired fibrinolysis may not only contribute to and aggravate the thrombotic events which determine the late complications of cardiovascular disease, but fibrin deposit also plays a role in the development of atherosclerotic lesions.¹¹

Fibrinolysis is an enzymatic system capable of dissolving clots. A proenzyme plasminogen can be converted to the active plasmin that will degrade fibrin by several types of activators, mainly pro-urokinase, urokinase and tissue plasminogen activator (t-PA). Inhibition of the system occurs at the level of activators and plasminogen activator inhibitor 1 (PAI 1) is the main plasminogen activator inhibitor. This 50 KD glycoprotein belongs to the super family of serine protease inhibitors (Serpin) and is found in endothelial cells, hepatocytes, platelets and plasma.

Recent results suggest that increased levels of PAI 1 are of pathogenetic significance and may be considered as risk factors for cardiovascular disease (review in Reference 12),

as well as for deep venous thrombosis in hip replacement or for fatal outcome in septicemia.^{13,14,15}

Circadian variations in the time of onset of myocardial infarct, sudden cardiac death and transient myocardial ischemia are well established with a peak incidence between 06.00 and 12.00.^{16,17} This is preceded for a few hours by a reduced fibrinolytic activity^{18,19} related to circadian variation in PAI 1 activity which peaks between 00.00 and 06.00.^{20,21} High plasma PAI 1 activity constitutes an independent risk factor for myocardial reinfarction in young subjects^{22,23} whose high PAI 1 activity correlates with reinfarction within 3 years but not with late reinfarction.²⁴

In this review we would like to describe the fibrinolytic abnormalities observed in the insulin resistance syndrome in cross sectional as well as in intervention studies, then to discuss the possible mechanisms leading to the main abnormality, i.e. increased PAI 1 levels.

PAI 1 and the insulin resistance syndrome

We ourselves and subsequently other investigators have demonstrated in various groups of patients, namely non diabetic subjects with various body weight,²⁵ premenopausal obese women,²⁶ subjects with angina pectoris,²⁷ and non insulin dependent diabetic patients^{8,7} that PAI levels were in strong correlation with body mass index, waist to hip girth ratio, plasma insulin and triglyceride which are hallmarks of insulin resistance syndrome (Table 1).²⁸ In a population study it was also shown that tissue plasminogen activator (tPA) and PAI were related to body mass index and waist to hip ratio in men and women and in women they were also related to systolic and diastolic

Table 1 Coefficient of correlation (r) between PAI 1 and insulinemia (ins), triglyceride (TG) and body mass index (BMI) in various groups of subjects

	PAI-1/Ins	PAI-1/TG	PAI-1/BMI
Normal subjects ²⁵	0.52***	0.16	0.66***
Obese non diabetic women ²⁶	0.72***	0.56***	0.50***
Non-insulin-dependent diabetic patients ⁷	0.60***	0.31	0.32*
Angina pectoris ²⁷	0.59***	0.25*	0.39**

blood pressure.²⁹

Multiple regression analysis has shown in most cases that these correlations disappeared after adjustment for insulin.^{7,25-27} However, a relationship subsisted between triglyceride and PAI 1 in pre-menopausal obese women.²⁶ These findings suggest that the link between plasma insulin and PAI 1 levels is independent of the other risk factors and that the relationship between PAI 1 level on the one hand and BMI, waist to hip circumference ratio and perhaps triglyceride on the other, could depend on the relationship between PAI 1 and insulin, this latter being a witness of the insulin-resistance state.

Insulin is accompanied in the plasma by insulin precursors and it has been questioned whether these precursors could not be the main associate of high PAI 1 levels.³¹ However, in non diabetic subjects these precursors do not exceed 10% of the total insulin-like molecules.³² With radioimmunoassay which we used for insulin determination, the cross-reactivity of these molecules with insulin to compete with insulin antibodies was only 25%. A contamination by these molecules would have only a limited effect on the close relationship generally observed between fasting insulin and PAI 1 levels in non diabetic patients.

The link between insulin resistance and PAI 1 levels observed in cross sectional studies was confirmed in intervention studies aimed at reducing insulin resistance. Firstly a simultaneous drop was shown in insulin, triglyceride and PAI 1 levels in 10 obese women after a 24 h fasting period.²⁵ In a 15 day double-blind placebo controlled study using Metformin (1.7 g/day) to reduce insulin resistance in non diabetic obese women, a decrease in fasting plasma insulin and triglyceride as well as PAI 1 levels was observed in the group treated but not in the placebo group.³³

Similar changes in PAI 1 have been reported after 2 week,³⁴ 6 week³⁵ or 13 week³⁶ hypocaloric diets. Interestingly Mehrabjan *et al.*³⁵ have shown that individuals with the highest initial PAI 1 and triglyceride levels were the most responsive to dietary measures (with a decrease of 45% and 34% respectively) whereas individuals with lowest levels exhibited no responsiveness.

The beneficial effect of physical exercise on insulin resistance is well known³⁷ and it has been shown that physical training also improved fibrinolytic activity.³⁸ It has recently been demonstrated that the increase in the activity after physical training was due to a decrease in PAI 1 levels in patients with myocardial infarct³⁹ and in sedentary men.⁴⁰

To summarize, it seems established that PAI 1 levels are elevated in patients who presented with the insulin resistance syndrome, namely in those who aggregate more or less excessive body weight, high waist to hip ratio, high blood pressure, hypertriglyceridemia and high plasma insulin levels. Among these subjects attempts at reducing insulin resistance are accompanied by a decrease in PAI 1 levels and amelioration of fibrinolytic activity in the majority of studies.

Attempt at explaining the elevated PAI levels in the insulin resistance syndrome

Curiously we have found PAI 1 levels in the normal range in normal weight non diabetic patients with Cushing's disease or acromegaly in spite of hyperinsulinemia and insulin resistance.⁴¹

The easiest way to study the mechanisms linking insulin resistance and PAI 1 levels would be to have an animal model with both insulin resistance and high PAI levels. The relationship between body weight, triglyceride and insulin on the one hand and PAI on the other led us to study two models of rats with nutritionally induced insulin resistance, one with excessive weight and the other one with hypertriglyceridemia.

Although rats do not develop atherothrombotic disease they do synthesize PAI 1 in various tissues such as heart, lung, liver.⁴² This synthesis may be stimulated by endotoxine, interleukin^{42,43} tumor necrosis factor⁴⁴ or lipopolysaccharide.⁴⁵ Elevation of plasma triglyceride by lipid rich diet is accompanied by modification of the fibrinolytic components in the rat.⁴⁶ For these reasons, we looked at the PAI levels in two models of induced insulin resistance in rats, one associated with excessive weight (the cafeteria fed rat), the other one induced by a fructose rich diet. In these animals, insulin resistance induces a compensatory hyperinsulinemia to maintain normal blood glucose levels and, at least in the second model, is associated with hypertriglyceridemia. Unfortunately neither of these two groups of rats had abnormal PAI 1 levels.⁴⁷

The close correlation between PAI levels on the one hand and triglyceride and insulin levels on the other suggests that the two latter could be candidates for triggering excessive secretion of PAI.

Stiko-Rham *et al.*⁴⁸ have shown that purified VLDL from hypertriglyceridemic patients increased PAI 1 production by endothelial cells. This enhancement was dose dependent with 100 µg/ml of these VLDL, giving rise to a 79% increase in PAI 1 secretion as compared with control cultures after 48-h incubation. VLDL from normolipemic patients induced a lower increase in PAI 1 production. This finding was confirmed by Mussoni *et al.*⁴⁹ In addition LDL and modified LDL may also stimulate PAI synthesis by endothelial cells.^{49,50,51} In the group of subjects with obesity, type 2 diabetes or coronary disease, high triglyceride levels are generally present among the other symptoms of the insulin resistance syndrome. To clarify the possible role

of triglyceride itself we compared PAI levels in three groups of hypertriglyceridemic patients: one with obesity, one of strictly normal body weight and one with alcohol induced hypertriglyceridemia.⁵² Overall, mean PAI levels in hypertriglyceridemic patients were higher than in controls. Compared to controls, both PAI activity and plasma insulin were elevated in obese hypertriglyceridemic patients, but not different from normal in the normal weight hypertriglyceridemic patients. In alcoholic hypertriglyceridemic patients, PAI activity was elevated but insulin was normal. In the whole series PAI activity was weakly correlated with triglyceride level ($r = 0.295$ $P < 0.05$), but in strong positive relationship with insulin levels ($r = 0.572$ $P < 0.001$). The lack of consistent association between triglyceride levels and PAI activity in these highly selected groups of patients does not plead for high triglyceride levels as the main cause of high PAI levels in subjects with insulin resistance.

Acute elevation of endogenous insulin by means of an oral glucose load given in the morning did not increase PAI levels nor break down the well known circadian rhythm of PAI levels⁵³ which shows a peak in the early morning and a nadir in the afternoon.^{20,21} Acute administration of exogenous insulin was ineffective.^{54,55} In the same way, infusion of insulin for 24 h, combined with glucose infusion to maintain normoglycaemia, does not modify this circadian rhythm.⁵³

PAI 1 concentration in the plasma is therefore not directly related to insulin level, at least under acute conditions which are not accompanied by insulin resistance or sustained non esterified fatty acid flux.

However, *in vitro* data suggest a direct effect of insulin to stimulate PAI synthesis and release.

Porcine endothelial cells in a culture stimulated by insulin, proinsulin or IGF1, respond by increased PAI 1 secretion.⁵⁶

When the hepatoma cell line Hep G2 is stimulated by insulin, at concentrations similar to those found in human portal vein a dose dependent increase in PAI release is observed^{57,58} accompanied by an enhanced PAI 1 mRNA expression.^{58,59} Similar results have been observed with stimulation by IGF1.⁵⁸ Interestingly, when the cells are made insulin resistant the stimulatory effect of insulin on PAI synthesis is enhanced. Preincubation with high dose

(10^{-7} M) of insulin with a view to reducing the number of insulin receptors by the mechanism of down regulation was followed by a subsequent stimulation with physiological (10^{-9} M) insulin concentration.⁵⁹ The decrease in insulin receptor was accompanied as expected by a reduction in (14 C) acetate incorporation, an index of lipid synthesis, whereas PAI 1 secretion and PAI 1 mRNA expression were enhanced. The addition of Metformin did not modify the effect of insulin on insulin receptors or (14 C) acetate incorporation. In contrast, the drug (10^{-4} M) inhibited insulin-mediated PAI 1 synthesis. These results indicate that PAI 1 synthesis in presence of insulin is markedly increased in down-regulated cells, and that Metformin inhibits this effect by acting at the cellular level. These *in vitro* data are relevant with regard to those found *in vivo* in insulin-resistant patients.

Taken together these *in vitro* data strongly indicate that PAI 1 synthesis in both hepatocytes and endothelial cells could be affected by high insulin concentration and lipoprotein abnormalities in patients with insulin-resistance states.

Although increased PAI 1 levels can now be considered as a part of the insulin resistance syndrome together with elevated waist to hip ratio and blood pressure on the clinical side and hyperinsulinemia, hypertriglyceridemia and low HDL cholesterol on the laboratory side, the mechanism of its raising remains poorly understood. Excess insulin, insulin intermediates and VLDL are candidates for triggering it, but no definite answer can be given at present.

Conclusion

Insulin resistance may increase the risk of coronary disease by promoting atheromatous lesions and development of thrombosis. A defect in the fibrinolytic system could be responsible for the initiation and the late complications of atherothrombosis. The increase in PAI 1 concentration linked to hyperinsulinaemia is a tempting partial explanation for the association between insulin resistance and coronary disease.

Normalization of PAI 1 by reducing insulin resistance could slow down atherothrombosis and prevent coronary disease. Further epidemiologic and intervention data are needed to confirm this hypothesis.

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